Sales Performance Analysis

A Narrative EDA of Sample Sales Data (2003–May 2005)

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Dataset	sample_sales_data
Period	January 2003 – May 2005
Total sales value	$10,030K across all records
Total orders	~99K order lines
Tools used	Power BI, Excel
Key dimensions	Sales, Quantity, Date, Customer, Product, Location, Order Status

The Headline

Between 2003 and mid-2005, this business grew consistently — average monthly sales rose from $293K in 2003 to $394K in 2004, and 2005 is tracking ahead of that pace despite missing its peak quarter.

But underneath that growth, three structural vulnerabilities deserve attention: customer concentration, product dependence, and a discount strategy that may be compressing margins on the deals that matter most.

Growth Trends: Consistent but Cyclical

Sales grew year-over-year across all three periods in the dataset. Monthly averages increased from $293K in 2003 to $394K in 2004, and 2005, with only five months of data and its peak quarter excluded, is running just $35K per month below 2004's full-year average. That is a strong signal of continuing momentum.

Figure 1; Sales Trend

However, the shape of that growth matters. Sales peak sharply in Q4 each year, then drop in Q1. The trend is not a smooth upward line; it is a repeating cycle of surge and retreat, with each cycle's peak slightly higher than the last. This indicates that the business lives and dies by Q4.

Growth: Customer and Product Concentration

The answer to what drives Q4 is also what should concern the business most. Three product lines: Classic Cars, Vintage Cars, and Motorcycles, account for 67.35% of all orders. Classic Cars alone represent 34.31%. These are not broad-market categories; they are niche collector products with a concentrated buyer base.

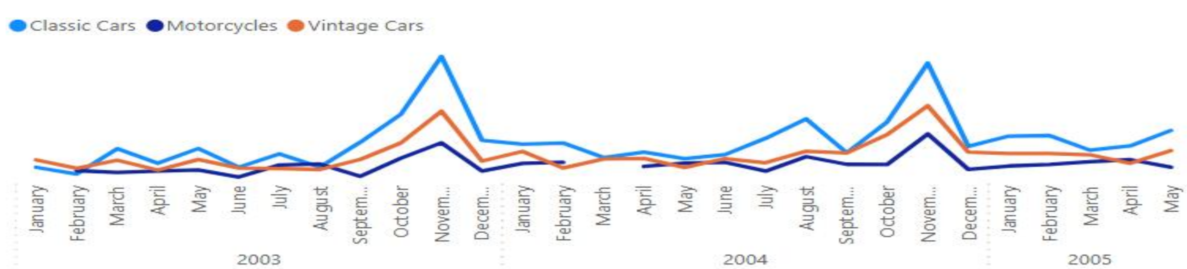

Figure 2; Top Products Order Trend

The customer picture compounds the problem. Out of 92 customers on record, the top 5 generate 21.31% of total revenue. Euro Shopping Channel alone accounts for $0.91M — nearly the size of the next two largest customers combined.

Figure 3: The top five customer and their sales value.

Key risk: The Q4 spike is not a broad market phenomenon.

It is driven primarily by a small number of high-value customers ordering from a narrow product range. If Euro Shopping Channel reduces orders, or if Classic Cars demand softens, the Q4 peak that defines this business's annual performance is directly at risk.

Geography Risk

The customer and product concentration problem extends into geography. The USA alone accounts for 36.16% of total sales. The USA, Spain, and France together drive 59.35%, meaning more than half of all revenue originates from three countries.

Figure 5Country Contribution in Sales

This is not just an interesting demographic fact. It is a risk amplifier. A regulatory shift, a logistics disruption, a currency move, or an economic downturn in any of these three markets does not hit the business proportionally; it hits it disproportionately hard.

This reveals that this business's growth rests on a narrow foundation.

Operations Performance

Not everything points to vulnerability. The operational picture is strong. 92.26% of all orders have been shipped. Dispute and cancellation rates are minimal — 0.6% disputed, 2.06% cancelled — and 1.68% of disputes were resolved. These are healthy numbers for a business of this order volume.

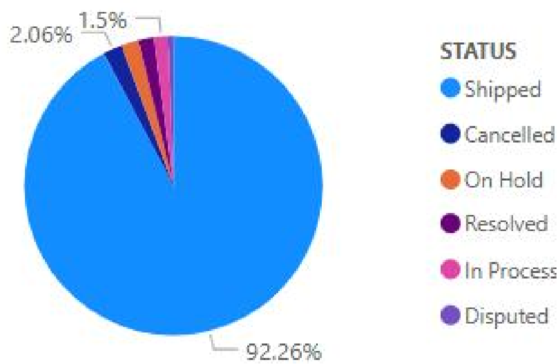

Figure 6Distribution of Total Orders Made by Status

This matters because it narrows the problem space. The supply chain is not the vulnerability. Fulfilment is not the risk. The risk is entirely on the demand side, in the concentration of who is buying, what they are buying, and where they are.

Pricing and Discount Strategy

Average discounts scale with deal size: 5.79% on small deals, 10.84% on medium deals, and 28.27% on large deals. On the surface, this looks like a rational volume-based pricing strategy. But read alongside the concentration findings above, it reveals a deeper problem.

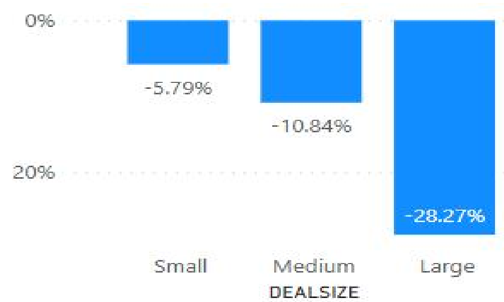

Figure 7; Average Discount on Deal Size

The customers receiving 28% average discounts are almost certainly the same top-5 customers generating 21% of total revenue. The business is most dependent on the customers it is also discounting most aggressively. It is, in effect, subsidising its own concentration risk.

Without cost data, margin erosion cannot be confirmed. But the pattern warrants scrutiny.

Volume-based discounting that protects market share with key accounts may be doing so at the expense of profitability on the orders that matter most. This is not an accusation; it is a hypothesis that the next layer of analysis should test.

Conclusion

The growth trend is genuine, consistent, and operationally well-supported. But sustaining that growth, and improving the quality of it, requires addressing three structural issues that the headline numbers obscure.

The business is growing, but is dangerously concentrated, and its biggest customers may also be its least profitable ones.

Recommendations

1. Review discount levels on large deals to understand whether volume discounts are margin-accretive or margin-destructive. This requires cost data not currently in the dataset.

2. Prioritise retention strategies for the top 5 customers while actively reducing dependency; diversifying the customer base so that no single account exceeds 5% of revenue.

3. Explore underpenetrated geographies. With 59% of revenue from three countries, there is significant room to grow in markets where the business currently has minimal presence.

Data Preparation Notes

No errors were identified in the source data. The following preparation steps were applied before analysis:

- Assigned appropriate data types to all columns
- Removed columns not relevant to this analysis: Phone, Address lines, Postal Code, Territory, Contact Names
- Engineered a PriceDifference% column: (PriceEach – MSRP) / MSRP, to capture pricing deviation from recommended retail